Explicit

Implicit

implicit methods are much easier to generalize

Forward Euler Method

Backward/Implicit Euler Method

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Spoiler: will use same principle for *distributional* grad flows

Discretizing Gradient Flows

- There's two main ways to discretize the gradient flow $\dot{x}(t) = -\nabla f(x(t))$:
  - **Explicit**: relying on direct computation of ∇f. E.g., Euler's method:

$$x_{t+1} = x_t - \tau \nabla f(x_t)$$

Forward Euler Method

  - **Implicit**: defining steps as solution to an optimization problem. E.g.,

$$x_{t+1} = \min_x \; f(x) + \frac{1}{2\tau} \|x - x_t\|^2$$

Backward/Implicit Euler Method

- Both share the same principle: make local steps that balance:

  i. reducing f

  ii. staying close to current location x_t

Spoiler: will use same principle for *distributional* grad flows

- But **implicit methods are much easier to generalize** (eg to non-Euclidean settings, ∇f not avaialable/expensive, etc) and more stable in practice.

From $\mathbb{R}^d$ to $\mathscr{P}(\mathbb{R}^d)$